POWER VS WEIGHT

A BUGFIX FOR “VETO DISTORTION”

with Jakob de Raaij, Ariel Procaccia, Jamie Tucker-Foltz

WEIGHTED VOTING GAMES

If you have five "players" voting with weights 5-1-1-1-1 by simple majority, the first player is *actually* a dictator (5x the weight of others, but all the power)

Likewise, five players voting with weights 8-8-8-8-1 by simple majority *actually* all have equal power

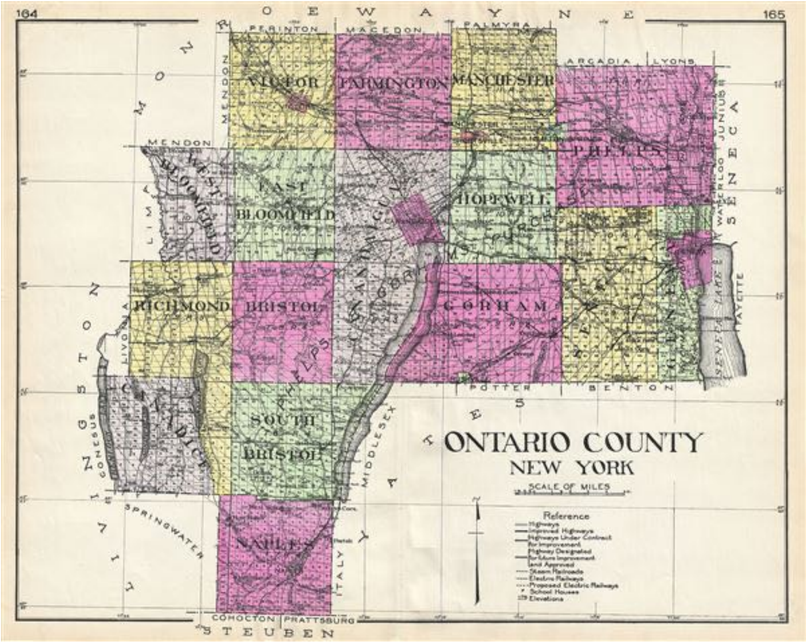
so vote weight is an illusion!

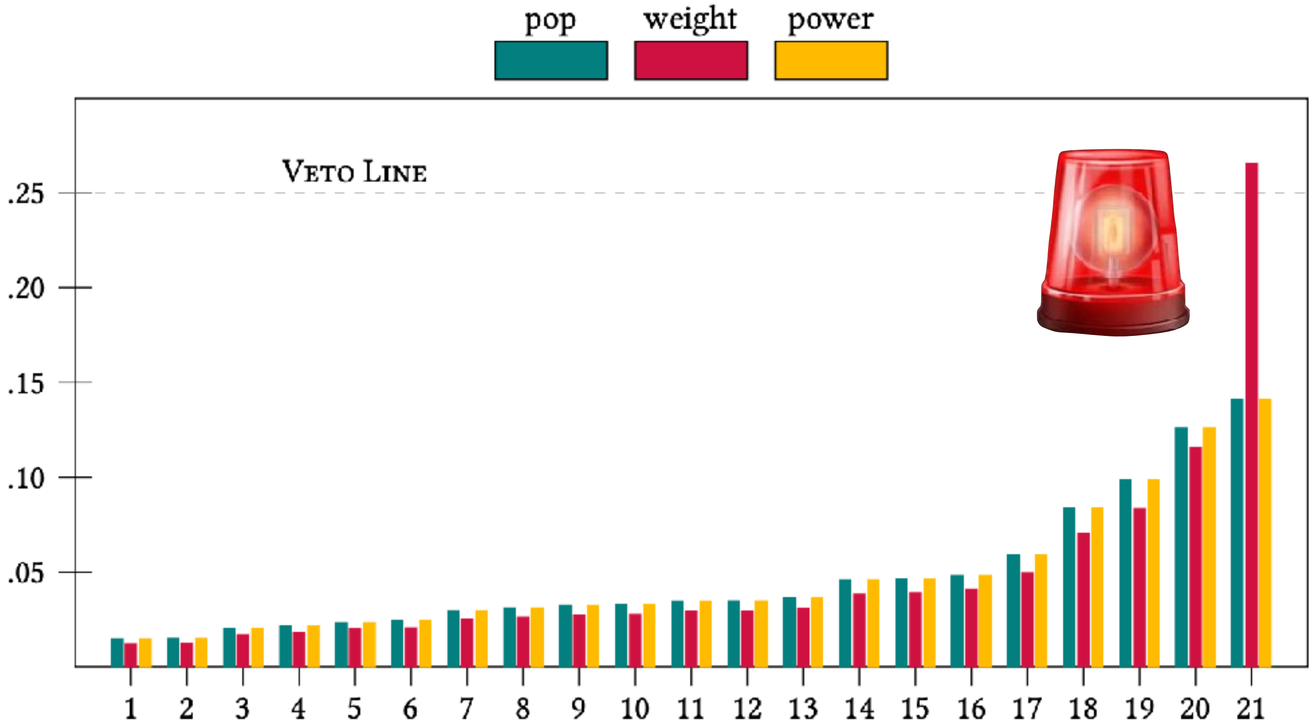
Let $\widehat{\beta}_i := \sum_{S \subset \mathscr{V} \setminus i} \sigma(S \cup i) - \sigma(S)$**, and let** β_i **be normalized to sum to 1.**

indicator for vote success

So power is your share of pivotality

Many NY counties have been (heuristically) solving the inverse problem for decades

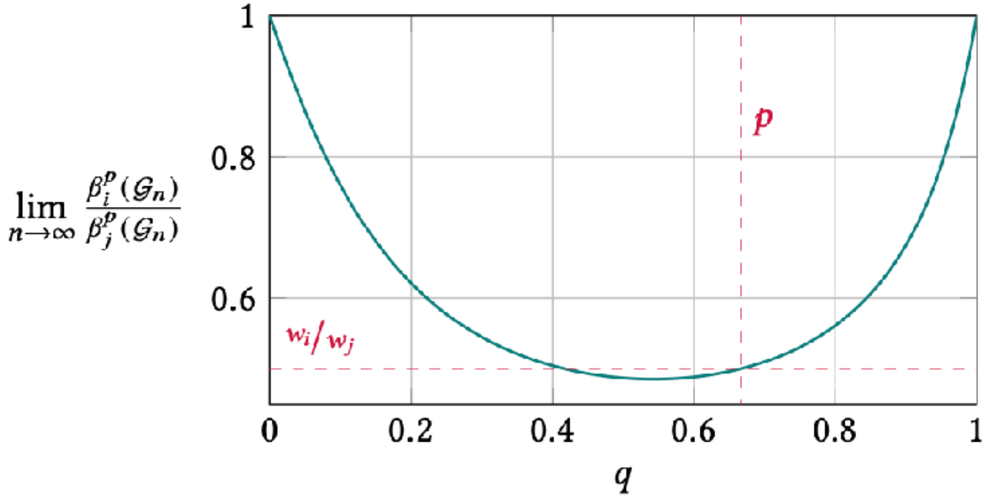


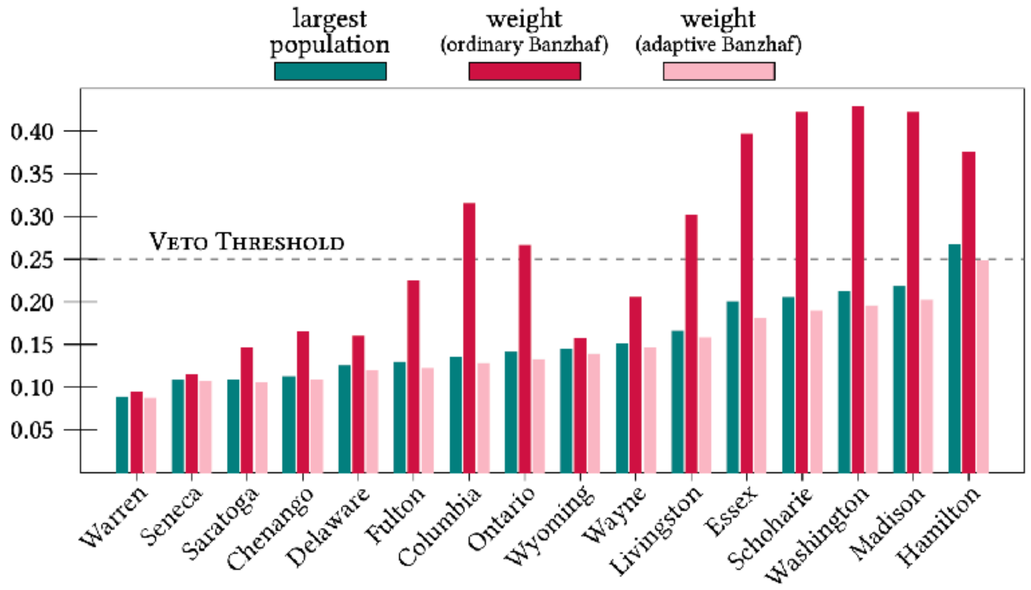


Simple intervention (dR-D-P-TF): use a behavioral model where each player has "propensity" q of voting for a matter that has threshold q for passage — so 3/4 chance of voting for something that needs 3/4 support to pass

Call this Adaptive Banzhaf

<u>Theorem</u>: for p-propensity voting, power ratios tend to weight ratios iff p=q





Better theorem: $14 \approx \infty$

MERCI

Ontario County	Town name	Adjusted pop.	Pop. share	$T = 1/2$ weights		$T = 2/3$ weights		$T = 3/4$
				CGR 2022	MGGG	CGR 2022	MGGG	MGGG
Town 1	Town of South Bristol	1641	0.01459	68	1487	68	1438	1229
Town 2	Town of Canadice	1668	0.01483	69	1511	69	1460	1247
Town 3	Town of Bristol	2284	0.02031	96	2105	95	1999	1711
Town 4	Town of Naples	2403	0.02137	100	2210	100	2104	1802
Town 5	Town of Seneca	2644	0.02351	110	2421	110	2313	1987
Town 6	Town of West Bloomfield	2740	0.02437	114	2505	114	2397	2058
Town 7	Town of Richmond	3360	0.02988	142	3123	140	2939	2522
Town 8	Town of Geneva	3473	0.03088	146	3203	144	3042	2607
Town 9	Town of East Bloomfield	3640	0.03237	152	3340	151	3192	2732
Town 10	City of Geneva (5,6)	3679	0.03272	153	3372	153	3223	2760
Town 11	City of Geneva (3,4)	3921	0.03487	163	3582	163	3435	2939
Town 12	Town of Hopewell	3931	0.03496	163	3588	163	3442	2948
Town 13	Town of Gorham	4106	0.03651	170	3741	171	3599	3080
Town 14	City of Canandaigua (2,3)	5140	0.04571	213	4684	214	4505	3855
Town 15	City of Geneva (1,2)	5210	0.04633	216	4741	217	4565	3903
Town 16	City of Canandaigua (1,4)	5436	0.04834	225	4932	226	4767	4076
Town 17	Town of Phelps	6637	0.05902	273	6001	277	5823	4985
Town 18	Town of Manchester	9404	0.08362	381	8381	394	8318	7045
Town 19	Town of Canandaigua	11109	0.0987	444	9750	464	9808	8339
Town 20	Town of Farmington	14170	0.12600	550	12093	601	12710	11576
Town 21	Town of Victor	15860	0.14103	602	13230	711	14921	26599
SUM		112,456		4550	100,000	4745	100,000	100,000
L1 error (discrepancy)				0.00103	0.00013	0.00207	0.00008	0.00012
				7.9× improved		26.3× improved		

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not close!